Project Proposal: Advanced Dimensionality Reduction and Feature Engineering for High-Performance Cancer Classification

This project focuses on applying a hierarchical progression of Dimensionality Reduction techniques—ranging from linear unsupervised methods to non-linear manifold learning—combined with biologically inspired feature engineering to enable efficient and highly accurate multi-class cancer classification.

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1. Background

Gene expression datasets often contain tens of thousands of gene features per sample. Training machine learning models on such high-dimensional data is computationally expensive and suffers from the "curse of dimensionality," where sparse data leads to overfitting. While classical techniques like Principal Component Analysis (PCA) provide a baseline, they often fail to capture complex, non-linear biological relationships. This project proposes a **gradual progression** workflow, moving from basic linear reduction to advanced supervised and non-linear techniques to maximize classification accuracy.[6]

2. Dataset

We will use the **UCI Gene Expression Cancer RNA-Seq dataset**, containing 801 tumor samples with 20,531 gene expression features. Each sample corresponds to one of several cancer types (e.g., BRCA, KIRC, LUAD).

3. Project Objectives

1. **Gradual Accuracy Progression:** Implement dimensionality reduction in three phases to demonstrate performance gains:
   - **Phase 1 (Baseline):** PCA (Unsupervised, Linear).
   - **Phase 2 (Intermediate):** LDA (Supervised, Linear).
   - **Phase 3 (Advanced):** t-SNE (Unsupervised, Non-Linear).
2. **Advanced Feature Engineering:** Incorporate domain-specific features to augment raw gene data:
   - **Pathway Activity Scores:** Aggregating genes into biological pathways (ssGSEA).
   - **Gene Pair Ratios:** Using ratios of specific gene pairs to create robust, self-normalizing features.
3. **Evaluation:** Compare the classification accuracy of SVM and Random Forest models across all three phases.

4. Methodology

4.1 Data Preprocessing

- **Standardization:** Log2 transformation followed by StandardScaler to normalize gene expression levels.
- **Feature Engineering:**
  - *Pathway Scoring:* We will map genes to KEGG/Reactome pathways and compute Single-Sample Gene Set Enrichment Analysis (ssGSEA) scores to reduce thousands of genes into ~300 pathway features.
  - *Gene Pair Ratios:* We will select top-ranking gene pairs (e.g., Gene A > Gene B) to generate features invariant to normalization errors.

4.2 Dimensionality Reduction Pipeline (The Progression)

We will implement three distinct reduction strategies to show the "Accuracy Ladder":

- **Step 1: Principal Component Analysis (PCA) – *The Baseline***
  - We will project data onto orthogonal axes of maximum variance. This is expected to yield ~95% accuracy but may miss subtle subtype differences due to its unsupervised nature.
- **Step 2: Linear Discriminant Analysis (LDA) – *The Supervised Boost***
  - We will apply LDA to find axes that maximize the separation *between* cancer classes. This should outperform PCA by utilizing class labels, potentially reaching ~98% accuracy.
- **Step 3: t-Distributed Stochastic Neighbor Embedding (t-SNE) – *The Non-Linear Peak***
  - We will use Parametric t-SNE to capture non-linear manifold structures that linear methods miss. This is expected to provide the highest accuracy (approaching 99%) by resolving complex clusters.[6]

4.3 Classification & Validation

- **Models:** Support Vector Machine (SVM) and Random Forest Classifier.
- **Validation:** Stratified K-Fold Cross-Validation (k=5).
- **Metrics:** Accuracy, Precision, Recall, F1-Score, and Confusion Matrix.

5. Expected Outcomes & Benefits

- **Quantifiable Progress:** A clear demonstration of accuracy improvement from PCA → LDA → t-SNE.
- **Biological Relevance:** The use of Pathway Scores and Gene Pair Ratios will make the model more interpretable to biologists compared to raw gene counts.
- **High Performance:** We expect the final t-SNE + Feature Engineering model to achieve state-of-the-art classification results.

6. References

1 Jolliffe, I. T. (2002). Principal Component Analysis.
2 UCI Machine Learning Repository – Gene Expression RNA-Seq Dataset.
3 Poličar et al. (2024). openTSNE: a modular Python library for t-SNE dimensionality reduction.
4 Georgiou et al. (2025). Stage-specific gene pair ratios highlight genes and mechanisms related to Multiple Myeloma. Computational and Structural Biotechnology Journal.
5 Chuiqin et al. (2024). irGSEA: The integration of single-cell rank-based gene set enrichment analysis. Brief. Bioinform.

Works cited

1. Machine learning approach to identify significant genes and classify cancer types from RNA-seq data - PMC - NIH, accessed on November 27, 2025, https://pmc.ncbi.nlm.nih.gov/articles/PMC12547461/
2. Multi-Class Cancer Classification with PCA and SVM.pdf
3. Top 12 Dimensionality Reduction Techniques for Machine Learning - Encord, accessed on November 27, 2025, https://encord.com/blog/dimentionality-reduction-techniques-machine-learning/
4. Capturing discrete latent structures: choose LDs over PCs - PMC - NIH, accessed on November 27, 2025, https://pmc.ncbi.nlm.nih.gov/articles/PMC9748550/
5. Understanding Dimensionality Reduction Techniques: Elbow Method, LDA, t-SNE, Autoencoders, and PCA | by Shivansh Srivastava | Medium, accessed on November 27, 2025, https://medium.com/@srivastavashivansh8922/understanding-dimensionality-reduction-techniques-elbow-method-lda-t-sne-autoencoders-and-pca-9d9890279dbc
6. FeatPCA: A feature subspace based principal component analysis technique for enhancing clustering of single-cell RNA-seq data - arXiv, accessed on November 27, 2025, https://arxiv.org/html/2502.05647v1